

Shear-lift magnitude audit | Scope

Existing dynamics

Frozen FEM → Stokes self resistance → Normal lubrication → Nonpenetration → Velocity

Read-only audit

Saved trajectories + local velocity gradients → slip, Reynolds numbers and force scales

Candidate comparison

Rigid-sphere Saffman magnitude → drag / lubrication comparison → applicability checks

Stage boundary

No added force · No trajectory recomputation · No automatic model development

Candidate Saffman magnitude only. Not included in trajectory dynamics.